

CSE 715: Computer Graphics

Chapter 9 — Geometric Representation: Bezier, B-Spline, Hermite Curves

Syllabus-Mandated Theory + Solved Exercises

University of Chittagong — 7th Semester B.Sc. (Engg.) Examination

Source: Schaum's Outline of Computer Graphics (Xiang & Plastock), Chapter 9 — notation follows the book exactly ($H_i(x)$, $B_{i,n}(x)$, $BE_{k,n}(x)$, $P(t)$)

Why this document looks different from the other chapter solutions: Bezier/B-spline/Hermite curves have **0/5 hits** in the 2020–2024 past papers — confirmed by direct re-verification of every question in every paper this exam season. `Syllabus.txt` explicitly overrides the "skip if not in past papers" rule for this one topic: "*Chapter 9: Bezier Curve, solve problems, B-spline, Hermite Curve, exercise and solved questions (other topics only previous years questions if any).*" So instead of a past-paper year-by-year breakdown, this document gives the book's core definitions plus worked textbook-style exercises, matching what the syllabus asks for. **Given the exam is imminent and this is a 0-yield topic, treat this as a stretch-goal skim, not a priority revision target** — Z-Buffer, Cohen-Sutherland, and the other TIER 1 topics come first.

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Core Definitions

Hermite Cubic Polynomials

Given knots t_{i-1}, t_i, t_{i+1} , the Hermite cubic basis functions are

$$H_i(x) = \begin{cases} -\frac{2(x-t_{i-1})^3}{(t_i-t_{i-1})^3} + \frac{3(x-t_{i-1})^2}{(t_i-t_{i-1})^2} & t_{i-1} \leq x \leq t_i \\ -\frac{2(t_{i+1}-x)^3}{(t_{i+1}-t_i)^3} + \frac{3(t_{i+1}-x)^2}{(t_{i+1}-t_i)^2} & t_i \leq x \leq t_{i+1} \end{cases} \quad \bar{H}_i(x) = \begin{cases} \frac{(x-t_{i-1})^2(x-t_i)}{(t_i-t_{i-1})^2} & t_{i-1} \leq x \leq t_i \\ \frac{(x-t_i)(t_{i+1}-x)^2}{(t_{i+1}-t_i)^2} & t_i \leq x \leq t_{i+1} \end{cases}$$

Hermitian cubic interpolation: given data points $(x_0, y_0), \dots, (x_n, y_n)$ and prescribed derivative (tangent-slope) values $y'_0, \dots, y'_n$ at each point, with knots $t_i = x_i$:

$$H(x) = \sum_{i=0}^n [y_i H_i(x) + y'_i \bar{H}_i(x)]$$

This constructs a piecewise-cubic curve that passes exactly through every data point *and* has the prescribed tangent slope there.

B-Splines

For a knot set $t_0, t_1, t_2, \dots$, the n th-degree B-splines $B_{i,n}$ are defined **recursively**:

$$B_{i,0}(x) = \begin{cases} 1 & t_i \leq x \leq t_{i+1} \\ 0 & \text{otherwise} \end{cases} \quad B_{i,n}(x) = \frac{x-t_i}{t_{i+n}-t_i} B_{i,n-1}(x) + \frac{t_{i+n+1}-x}{t_{i+n+1}-t_{i+1}} B_{i+1,n-1}(x)$$

$B_{i,n}(x)$ is nonzero only on $[t_i, t_{i+n+1}]$ (**local support** — the key advantage over Bernstein/Bezier). For nonrepeated knots, $B_{i,n}(t_i) = 0 = B_{i,n}(t_{i+n+1})$ for $n \geq 1$. Repeated knots are allowed (using the convention $\frac{0}{0} = 0$). **Spline of degree m :** an m th-degree spline through $n+1$ data points is $S_m(x) = \sum_{i=0}^{m+n-1} a_i B_{i,m}(x)$. For the **cubic spline** ($m=3$): $n+3$ coefficients a_i need $n+3$ equations — $n+1$ come from the interpolation criterion $S_3(x_j) = y_j$, and the remaining 2 from a chosen **boundary condition**: natural ($S_3''(x_0) = S_3''(x_n) = 0$), clamped (prescribed end slopes), cyclic (for closed curves), or anticyclic.

Bernstein Polynomials

Degree- n Bernstein polynomials over $[0, 1]$:

$$BE_{k,n}(x) = \frac{n!}{k!(n-k)!} x^k (1-x)^{n-k}, \quad 0 \leq x \leq 1$$

Cubic ($n = 3$) Bernstein basis: $BE_{0,3} = 1 - 3x + 3x^2 - x^3$, $BE_{1,3} = 3(x - 2x^2 + x^3)$, $BE_{2,3} = 3(x^2 - x^3)$, $BE_{3,3} = x^3$.

Bezier-Bernstein Approximation (the "Bezier Curve")

Given a guiding polyline (control points) $P_0, \dots, P_n$, the **Bezier curve** is

$$P(t) : \begin{cases} x(t) = \sum_{i=0}^n x_i BE_{i,n}(t) \\ y(t) = \sum_{i=0}^n y_i BE_{i,n}(t) \\ z(t) = \sum_{i=0}^n z_i BE_{i,n}(t) \end{cases} \quad 0 \leq t \leq 1$$

Four properties (frequently asked as "explain properties of Bezier curve"):

1. **Endpoint interpolation:** $P_0 = P(0)$, $P_n = P(1)$ — the curve touches the first and last control points exactly (but not the interior ones, in general).
2. **Tangent direction at the endpoints** matches the guiding polyline's first/last segment: $P'(0) = n \cdot (P_1 - P_0)$, $P'(1) = n \cdot (P_n - P_{n-1})$.
3. **Convex hull property:** the entire Bezier curve lies within the convex hull (the "rubber band" polygon) of the control points $P_0, \dots, P_n$.
4. **Suited to interactive/piecewise design:** multiple Bezier curves can be joined with continuous tangents by keeping the two adjacent guiding-polyline edges at the shared endpoint collinear.

Bezier-B-Spline Approximation

Replaces Bernstein polynomials with B-splines for **local control**:

$$P(t) : \begin{cases} x(t) = \sum_{i=0}^n x_i B_{i,m}(t) \\ y(t) = \sum_{i=0}^n y_i B_{i,m}(t) \\ z(t) = \sum_{i=0}^n z_i B_{i,m}(t) \end{cases} \quad 0 \leq t \leq n - m + 1$$

Five properties:

1. Shares all Bezier-Bernstein properties (endpoint interpolation, matching end-tangents, convex hull) — in fact identical to it when $m = n$.
2. If $m + 1$ consecutive control points are collinear, that span of the curve is linear.
3. **Local control:** moving one control point only changes $m + 1$ nearby spans, not the whole curve (unlike Bezier-Bernstein, where moving any control point can reshape the entire curve).
4. Fits the guiding polygon more closely than Bezier-Bernstein.
5. Control points can be given **multiplicity** $k \geq 2$ — with multiplicity $m + 1$, the curve is pulled all the way through that control point.

Cavalier/cabinet-style memory hook: Bezier-Bernstein = "one control point moves, whole curve reshapes" (global); Bezier-B-spline = "one control point moves,

only nearby span reshapes” (local) — this local-vs-global contrast is the single most quotable fact if only one property can be remembered.

Solved Exercises (adapted from Schaum’s Solved Problems 9.3, 9.4, 9.6)

Exercise 1 — B-spline is Zero Outside Its Support Interval

Show that $B_{i,n}(x) = 0$ if $x < t_i$ or $x > t_{i+n+1}$.

By induction on the degree n , base case $n = 0$: illustrate the first step for degree 1, $B_{i,1}(x)$. Suppose $x < t_i$; then also $x < t_{i+1}$, so both zero-degree splines $B_{i,0}(x) = 0$ and $B_{i+1,0}(x) = 0$ (since x is left of both their support intervals). By the recursive formula, $B_{i,1}(x)$ is a combination of $B_{i,0}(x)$ and $B_{i+1,0}(x)$, both zero, so $B_{i,1}(x) = 0$. For the boundary case $x = t_i$ (nonrepeated knots): $B_{i,0}(t_i) = 1$ but $B_{i+1,0}(t_i) = 0$, giving

$$B_{i,1}(t_i) = \frac{t_i - t_i}{t_{i+1} - t_i}(1) + \frac{t_{i+2} - t_i}{t_{i+2} - t_{i+1}}(0) = 0$$

Similar reasoning handles $x \geq t_{i+2}$. The same induction pattern extends to any degree n : each step only ever combines two lower-degree splines that are individually zero outside their own (narrower) support, so the zero-outside-support property propagates upward.

Exercise 2 — Knot Set for Cubic B-Spline Interpolation

Given data points $P_0(0, 0), P_1(1, 2), P_2(2, 1), P_3(3, -1), P_4(4, 10), P_5(5, 5)$, find a knot set $t_0, \dots, t_9$ for cubic B-spline interpolation ($m = 3, n = 5$).

Need $m + n + 2 = 10$ knots $t_0, \dots, t_9$. **Scheme:** repeat the boundary knots $m + 1 = 4$ times at each end:

$$t_0 = t_1 = t_2 = t_3 = -1 (< x_0 = 0) \quad t_6 = t_7 = t_8 = t_9 = 6 (> x_5 = 5)$$

Remaining interior knots via $t_{i+4} = \frac{x_{i+1} + x_{i+2} + x_{i+3}}{3}$ for $i = 0, 1$ (the averaging scheme) or the simpler alternative $t_{i+4} = x_{i+2}$:

$$t_4 = x_2 = 2, \quad t_5 = x_3 = 3$$

Final knot set: $(-1, -1, -1, -1, 2, 3, 6, 6, 6, 6)$.

Exercise 3 — Bezier Curve: Endpoint and Midpoint Evaluation

Illustrative exercise (not from a past paper) showing the standard "evaluate a cubic Bezier curve at a given t " computation — a natural professor question given the syllabus explicitly asks for "solve problems" on Bezier curves.

Given cubic Bezier control points $P_0(0,0)$, $P_1(1,3)$, $P_2(3,3)$, $P_3(4,0)$, find the curve point at $t = 0.5$.

Using the cubic Bernstein basis from §1.3: at $t = 0.5$, $BE_{0,3} = 1 - 3(.5) + 3(.25) - .125 = 0.125$, $BE_{1,3} = 3(.5 - 2(.25) + .125) = 3(.125) = 0.375$, $BE_{2,3} = 3(.25 - .125) = 3(.125) = 0.375$, $BE_{3,3} = .125$. (Check: sum = $0.125 + 0.375 + 0.375 + 0.125 = 1$ ✓.)

$$x(0.5) = 0(.125) + 1(.375) + 3(.375) + 4(.125) = 0 + 0.375 + 1.125 + 0.5 = 2.0$$

$$y(0.5) = 0(.125) + 3(.375) + 3(.375) + 0(.125) = 1.125 + 1.125 = 2.25$$

So $P(0.5) = (2.0, 2.25)$. **Sanity checks from the properties in §1.4:** $P(0) = P_0 = (0,0)$ ✓ (endpoint interpolation); the curve point at $t = 0.5$ lies inside the convex hull (the quadrilateral $P_0P_1P_2P_3$) ✓.

Quick Summary

Concept	One-line memory hook
Hermite cubic	Interpolates through points <i>and</i> matches prescribed tangent slopes there
B-spline (general)	Recursive basis, built up from degree 0; local support on $[t_i, t_{i+n+1}]$
Bernstein polynomial	$BE_{k,n}(x) = \binom{n}{k} x^k (1-x)^{n-k}$ — the basis functions behind Bezier
Bezier-Bernstein curve	Endpoints interpolated, tangents match polyline ends, convex hull, but global control
Bezier-B-spline curve	Same nice properties as Bezier, but local control + closer polygon fit + control-point multiplicity

Exam-day reality check: this topic has never appeared in 5 years of past papers. If it's the night before the exam and time is short, skip straight back to Z-Buffer, Cohen-Sutherland, Cavalier/Cabinet, and the other TIER 1/TIER 2 topics — come back here only if all of those are solid.